Explicit bounds and algorithms for torsion points

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based on joint work in progress with **Evelina Viada**

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Abelian varieties and their points

Fix an **abelian variety** A defined over a **number field** K. Let $g = \dim(A)$.

For any field $L \supseteq K$, we $A(L)_{\text{tors}}$ denote the **torsion subgroup** of $A(L)$.

Given a closed algebraic subvariety $V \subseteq A$, can we determine the **Zariski closure** of the intersection $V(\overline{K}) \cap A(\overline{K})_{\text{tors}}$, denoted by V_{tors}? Can we bound its **degree**?

For $V \subseteq \mathbb{P}^n$, let $\delta(V)$ the **minimal degree** of the sets of polynomials **generating** the defining ideal $I(V)$. Then $\delta(V) \leq \deg(V) \leq \delta(V)^{n-\dim(X)}$.

Shafarevic: If $\dim(A) = g$ there exists an embedding $A \subseteq \mathbb{P}^{2g+1}$. This is optimal (**Barth**, **van de Ven**).

Kempf: If $A \subseteq \mathbb{P}^n$ and $\nu_d \colon \mathbb{P}^n \hookrightarrow \mathbb{P}^{N_d}$ is the **Veronese embedding**, where we set $N_d = \binom{n+d}{d} - 1$, then $\delta(\nu_3(A)) \leq 3$ and $\delta(\nu_d(A)) = 2$ whenever $d \geq 4$.

For products of elliptic curves $A = E_1 \times \cdots \times E_g$ we use the **Segre embedding** $A \subseteq (\mathbb{P}^2)^g \subseteq \mathbb{P}^{3^g-1}$.

A conjecture by Manin and Mumford

A closed subvariety $X \subseteq A$ is **torsion** if $X = \bigcup_{s=1}^{S} B_s + T_s$ where $B_s \subseteq A$ is an abelian subvariety, and $T_s \in A(\overline{K})_{\text{tors}}$.

Manin, **Mumford**, **Lang**: For a closed subvariety $V \subseteq A$, is V_{tors} torsion?

Yes! Existing proofs use the **action of Galois** (**Raynaud**, **Hindry**, **Richard**, **Galateau & Martínez**), p-**adic methods** (**Coleman**, **Buium**, **Poonen**, **Miller & Morrow**), **model theory** (**Hrushovski**), **algebraic geometry** (**Pink & Rössler**), **formal groups** (**Boxall**, **Rössler**) or o-**minimality** (**Pila & Zannier**).

Given V, can we determine V_{tors} **algorithmically**?

Write $V_{\text{tors}} = \bigcup_{s=1}^{S(V)} B_s + T_s$. Can we **bound** $\deg(B_s)$ and $\operatorname{ord}(T_s)$, for every s?

Since $[K(T)\colon K] \geq (6g)^{-8} \max(1, h_F(A), \log((6g)^{-8} n^{1/g}))^{-1} \operatorname{ord}(T)^{1/g}$ by **Gaudron & Rémond**, bounding $S(V)$ is **equivalent** to bounding $\operatorname{ord}(T_s)$ for every s.

Bounding $S(V)$ and $\deg(B_s)$ is **equivalent** to bounding $\deg(V_{\text{tors}})$.

If these bounds are **computable** (or even **explicit**) and **small**, we can find V_{tors}!

Finding the torsion points

Input: Equations of $V \subseteq A$, and real numbers $X, Y > 0$ such that $\deg(B_s) \leq X$ and $\mathrm{ord}(T_s) \leq Y$, for any s.

Step 1: Compute $\phi\colon A \to A_1^{m_1} \times \cdots \times A_r^{m_r}$, using **isogeny estimates**.

Gaudron & Rémond: $\deg(\phi) \leq ((7g)^{8g^2}[K:\mathbb{Q}]\max(1, \log[K:\mathbb{Q}], h_F(A)))^{4g^2}$.

Suppose for simplicity that $r = 1$, and let $A = G^m$, where G is simple.

Step 2: Find all $B \subseteq A$ with $\deg(B) \leq X$, writing $B = M \cdot G^r$ for some $M \in \mathrm{End}(G)^{m \times r}$. Then $\deg(B)$ can be computed from $\mathrm{nrd}(M^\dagger \cdot M)$, where $\dagger$ is the **Rosati involution**.

Liebendörfer & Rémond have an explicit expression when $\mathrm{End}(G)$ is maximal.

Step 3: For each $T \in A(\overline{K})_{\mathrm{tors}}$ with $\mathrm{ord}(T) \leq Y$, and each B as above, check if $B \subseteq V - T$. This can be done efficiently on **tangent spaces**.

To run this algorithm, we need to keep X and Y **small**...

Bounding the degrees of maximal sub-translates

Given $V \subseteq A$, we want to bound $\deg(B+P)$ for $B+P \subseteq V$ maximal.

Bombieri & Zannier: $\deg(B+P) \leq \deg(V)^{c_1}$, where $c_1 = 2^{\dim(V)-\dim(B)}$.

This bound is **explicit** and **uniform**, but not too **small**...

For equidimensional $V \subseteq \mathbb{P}^n$, let $\delta_0(V)$ be the degree of **incomplete definition**.

P. & Viada: We have that $\deg(B+P) \leq c_2 \cdot \delta_0(V)^{g-\dim(B)}$, where we define $c_2 = 2^{((\dim(V)-\dim(B))^2-1)(g-1)} \deg(A)$.

To prove this, we can assume that $P = 0$ and that V is irreducible.

If V is an abelian subvariety, $B = V$ and $\deg(B) = \deg(V) \leq \deg(A)\delta_0(V)^{g-\dim(V)}$.

Choose hypersurfaces $F_1, \ldots, F_{g-\dim(V)} \subseteq A$ with $\deg(F_i) \leq \delta_0(V)$ and $V \subseteq^c \bigcap_i F_i$.

If V is not an abelian subvariety, then $B \subseteq V \cap [2] \cdot F_i \subsetneq V$ for some F_i.

Conclude by **induction** and **Bézout's theorem**, using $\deg([2]F_i) = 2^{2(g-1)} \deg(F_i)$.

Quantitative and uniform versions of Manin-Mumford

Buium, **Poonen**: If $K = \mathbb{Q}$ and $\dim(V) = 1$, while $A = \mathrm{Jac}(V)$ and $g = \dim(A)$, for any prime p of good reduction for V we have that $S(V) \leq 2p^{3g}3^g(p(g-1)+3g)$.

If $A \sim A_1^{m_1} \times \cdots \times A_r^{m_r}$, let $d = \dim(A_1) + \cdots + \dim(A_r)$ and $g = \dim(A)$. Fix $\mathfrak{p} \subseteq \mathcal{O}_K$ of good reduction for A. Let $q = \#(\mathcal{O}_K/\mathfrak{p})$, and $A_+ = \{(x,y,z) \in A^3 : x+y=z\}$.

Hrushovski: $S(V) \leq \deg(A_+)^{c_3} \cdot \deg(V)^{c_4}$, where $c_3 = 2^9(d+1)^7 \log_2^2(1+\sqrt{q})$, while $c_4 = 2^{16(d+1)^3 g}$.

Gao & Ge & Kühne: $S(V)$ can be bounded solely in terms of $\dim(A)$ and $\deg(V)$.

David & Philippon: If $A = E^g$ is a power of an elliptic curve, then we have that $S(V) \leq \deg(V)^{c_5}$, where $c_5 = 7g^{4(\dim(V)+1)^2+2}$.

Yu & Yuan & Zhou: If $\dim(V) = 1$ and $A = \mathrm{Jac}(V)$, then $S(V) \leq 3.2 \cdot 10^{11} \cdot g^{\frac{17}{3}}$.

Looper & Silverman & Wilms: Let $K = \kappa(B)$ for some curve B over a field $\kappa = \overline{\kappa}$. If $\dim(V) = 1$ and $A = \mathrm{Jac}(V)$ then $S(V) \leq 16g^2 + 32g + 124$.

Explicit Manin-Mumford in abelian varieties

To find V_{tors}, we can use the **Galois theory** of $A(\overline{K})_{\text{tors}}$.

Serre, **Wintenberger**: There exists an integer $\lambda \geq 1$ such that for each $N, M \in \mathbb{N}$ which are coprime there exists $\sigma \in \text{Gal}(\overline{K}/K)$ which acts on $A(\overline{K})[N] \cong (\mathbb{Z}/N\mathbb{Z})^{2g}$ as multiplication by M^{λ}. Let $\lambda(A)$ be the minimal such integer.

For each j, let V_{tors}^{j} let the j-equidimensional part of V_{tors}.

Galateau & Martínez: Fix $A \subseteq \mathbb{P}^n$ and let $g = \dim(A)$. Then, one has that $\deg(V_{\text{tors}}^{j}) \leq c_5 \cdot \delta(V)^{g-j}$ where $c_5 := (16^{g(\lambda(A)+2)} n)^{(g-j)\dim(V)} \deg(A)$.

This allows one to compute V_{tors}, **when one knows** $\lambda(A)$.

Galateau & Martínez construct $V' \subseteq A$ with $V_{\text{tors}} \subseteq V \cap V' \subsetneq V$, obtained as a union of torsion translates of Galois conjugates of V. Hence $\delta(V')$ is controlled by $\delta(V)$.

They conclude using a **geometric induction** due to **Amoroso & Viada**.

Galateau & Martínez: If $\dim(V) = 1$, then $S(V) \leq 4^{(2\lambda(A)+1)g} \deg(V)^2$.

Galois representations attached to abelian varieties

Let K be number field and $A_{/K}$ be an abelian variety with $g = \dim(A)$. Then, we have the Galois representation $\rho_A \colon \Gamma_K := \mathrm{Gal}(\overline{K}/K) \to \mathrm{Aut}_{\mathbb{Z}}(A(\overline{K})_{\mathrm{tors}}) \cong \mathrm{GL}_{2g}(\widehat{\mathbb{Z}})$ where $\widehat{\mathbb{Z}} := \varprojlim_N \mathbb{Z}/N\mathbb{Z} = \prod_p \mathbb{Z}_p$ with $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$.

Serre, **Wintenberger**: There exists $\lambda(A) > 0$ such that $(\widehat{\mathbb{Z}}^\times)^{\lambda(A)} \subseteq \rho_A(\Gamma_K)$.

Serre: If $\mathrm{MT}(A)$ denotes the integral **Mumford-Tate group** of A then $\rho_A(\Gamma_K)$ **should be** commensurable with $\mathrm{MT}(A)(\widehat{\mathbb{Z}})$ whenever $\mathrm{MT}(A)$ is **maximal**.

For a prime ℓ, let $\rho_{A,\ell^\infty} \colon \Gamma_K \to \mathrm{Aut}_{\mathbb{Z}}(A(\overline{K})[\ell^\infty]) \cong \mathrm{GL}_{2g}(\mathbb{Z}_\ell)$.

Cadoret & Moonen: If $\mathrm{MT}(A)_{\mathbb{Q}_\ell} = (\overline{\rho_{A,\ell^\infty}(\Gamma_K)})^0_{\mathbb{Q}_\ell}$ for some prime ℓ, the previous conjecture holds true. Then, $\rho_A(\Gamma_K) \cap \widehat{\mathbb{Z}}^\times$ has finite index inside $\widehat{\mathbb{Z}}^\times$.

Eckstein: If A has complex multiplication, *i.e.* $\mathrm{End}(A) \otimes \mathbb{Q} \supseteq F$ where F is a number field of degree $2g$, then $\lambda(A) \leq [K \colon \mathbb{Q}] \cdot \#\mathrm{GL}_{2g}(\mathbb{Z}/3\mathbb{Z}) \leq [K \colon \mathbb{Q}] \cdot 3^{4g^2}$.

Galateau & Martínez worked towards making effective the computation of $\lambda(A)$.

Galois representations attached to elliptic curves

Serre: If E is an elliptic curve defined over a number field K then we have that $\rho_E(\Gamma_K) \subseteq \mathrm{MT}(E)(\widehat{\mathbb{Z}})$ is open, *i.e.* closed and of finite index $\mathcal{I}(E/K)$.
If $\mathrm{End}(E_{\overline{K}}) \cong \mathbb{Z}$ then $\mathrm{MT}(E)(\widehat{\mathbb{Z}}) = \mathrm{GL}_2(\widehat{\mathbb{Z}})$. Otherwise $\mathrm{End}(E_{\overline{K}}) \cong \mathcal{O}$ for some order $\mathcal{O}$ inside an imaginary quadratic field F, and $|\mathrm{MT}(E)(\widehat{\mathbb{Z}}) \colon \widehat{\mathcal{O}}^\times| \leq 2$.

Campagna & P.: If $\mathrm{End}(E) \cong \mathcal{O}$ then $\lambda(E) = \mathcal{I}(E/K)$ can be computed from $K(E[3])$. If $\Delta_{\mathcal{O}} < -4$ and $KF \cap F^{\mathrm{ab}} = F(j(E))$, then $\lambda(E) \leq 2$.

Sutherland, **Zywina**: $\rho_E(\Gamma_K)$ can be computed algorithmically.

Serre: Is it true that $\mathcal{I}(E/K)$ can be bounded solely in terms of K? In particular, is it true that $\mathcal{I}(E/\mathbb{Q}) \leq 2736$ (**Zywina**)?

Lombardo: $\mathcal{I}(E/K) \leq \exp(1.9 \cdot 10^{10}) \cdot ([K \colon \mathbb{Q}] \cdot \max(1, h_F(E), \log[K \colon \mathbb{Q}]))^{12395}$.

Furio: $\mathcal{I}(E/\mathbb{Q}) \leq 2.5 \cdot 10^6 \cdot (51 N_E (1 + \log\log(N_E))^{\frac{1}{2}})^{2\omega(N_E)}$ where $N_E = \mathrm{rad}(\mathfrak{f}_E)$.

Lombardo & Tronto: If E is defined over $\mathbb{Q}$ then $\max_\ell |\mathbb{Z}_\ell^\times \colon \mathbb{Z}_\ell^\times \cap \rho_{E,\ell^\infty}(\Gamma_{\mathbb{Q}})| \leq 36$.

Explicit families of torsion points on Serre curves

An elliptic curve $E_{/\mathbb{Q}}$ is a **Serre curve** if $\mathcal{I}(E/\mathbb{Q}) = 2$. In this case $\lambda(E) = 1$.

Jones: 100% of elliptic curves ordered by height are Serre curves.

Daniels: There exists an explicit thin set $S \subseteq \mathbb{Q}$ such that for each $t \in \mathbb{Q} \setminus S$ the elliptic curve $E_t \colon y^2 + xy = x^3 + t$ is a Serre curve.

P. & Viada: For each $t \in \mathbb{Q}$, and each $a \in \mathbb{Q}^\times$ let $C_{a,t} \colon \{y_2 = ax_1\} \subseteq E_t^2$. Then $(C_{a,t})_{\text{tors}} = \{0\} \times (E[2](\overline{\mathbb{Q}}) \setminus \{0\})$ for each $t \in \mathbb{Q} \setminus S$ and $a \in \mathbb{Q}^\times$.

Thanks to **Galateau & Martínez**, we have to compute around 240 division polynomials $\phi(x, t)$, and then check that the corresponding resultants cannot vanish if $a, t \in \mathbb{Q}$. This can be achieved through sieving methods.

To compare, in this case **Galateau & Martínez** give $\#(C_{a,t})_{\text{tors}} \leq 17775$, while **David & Philippon** give $\#(C_{a,t})_{\text{tors}} \leq 15^{1835008}$. Other results don't strictly apply to this case.

In fact, $-1 \in \rho_{E_t}(\Gamma_{\mathbb{Q}})$ and $C_{a,t} \neq -C_{a,t}$. Hence, $\#(C_{a,t})_{\text{tors}} \leq \deg(C_{a,t})^2 = 225$. Similar ideas are used by **Baker** and **Tamagawa**.

Poonen's family

Stoll: What is $\max(S(V))$ for $\dim(V) = 1$ of fixed genus $g \geq 2$, and $A = \mathrm{Jac}(V)$?

Fu & Stoll: This is intimately related to a conjecture by **Bogomolov & Fu & Tschinkel**, proven true by **de Marco & Krieger & Ye**, and later by **Poineau**.

Consider the family of elliptic curves $E_t\colon y^2 = x^3 + 3(2t-5)x + (t^2 - 14t + 22)$, and let $C_t = \{4x_1x_2 = (t+10)x_1 - (t-10)x_2 - 24\} \subseteq E_t \times E_{-t}$. Note that C_t has the plane model $y^2 = x^6 + tx^3 + 1$.

Poonen: There exists an infinite set $S \subseteq \overline{\mathbb{Q}} \cap \mathbb{R}$ such that $\#(C_\tau)_{\mathrm{tors}} \geq 22$ when $\tau \in S$. If $\tau^2 = 1300$ then $\#(C_\tau)_{\mathrm{tors}} = 34$, as found by **Stoll**.

Considering E_t as an elliptic curve over $\mathbb{Q}(t)$, one can compute $G := \rho_{E_t}(\Gamma_{\mathbb{Q}(t)})$. Using results of **Cadoret**, **Zywina** and **Litt**, one can uniformly bound $|G\colon \rho_{E_\tau}(\Gamma_{\mathbb{Q}(\tau)})|$, and thus $\mathcal{I}(E_\tau/\mathbb{Q}(\tau))$, for each τ such that $\tau^2 \in \mathbb{Q}$.

P. & Viada: For most τ with $\tau^2 \in \mathbb{Q}$, we have that $\#(C_\tau)_{\mathrm{tors}} = 10$. In general, we should have $\#(C_\tau)_{\mathrm{tors}} \leq 34$, as proven by careful analysis of $\rho_E(\Gamma_{\mathbb{Q}(\tau)})$.

Thank you very much for your attention!



It is perhaps the great discomfort
of those trying to silence the world
to discover that we have voices
sealed inside our heads,
voices that, with each passing day,
grow even louder
than the clamor of the world outside.

Edwige Danticat,
"The farming of Bones"